



ITM SKILLS
UNIVERSITY

Mental Health Case Tracking System

By Ashish SK Gupta

Index

1. Problem Statement
2. Solution
3. Working (Method/Mind map)
4. Conclusion.

Note: Slides (excluding Thank-you and Introduction slide) should not be more than 4.



Problem Statement

The Challenge

Mental health cases need confidential long-term tracking.

Paper records risk privacy leaks and are hard to manage.

Counsellors need a secure system to manage patient records and therapy sessions.

Objectives & Requirements

- Ensure confidential case management
- Case registration with unique IDs
- Session tracking per patient
- Secure login / password protection
- Persistent & secure data storage

Solution

Login / Register

FNV-1a password hashing
Role-based access control
Secure account creation

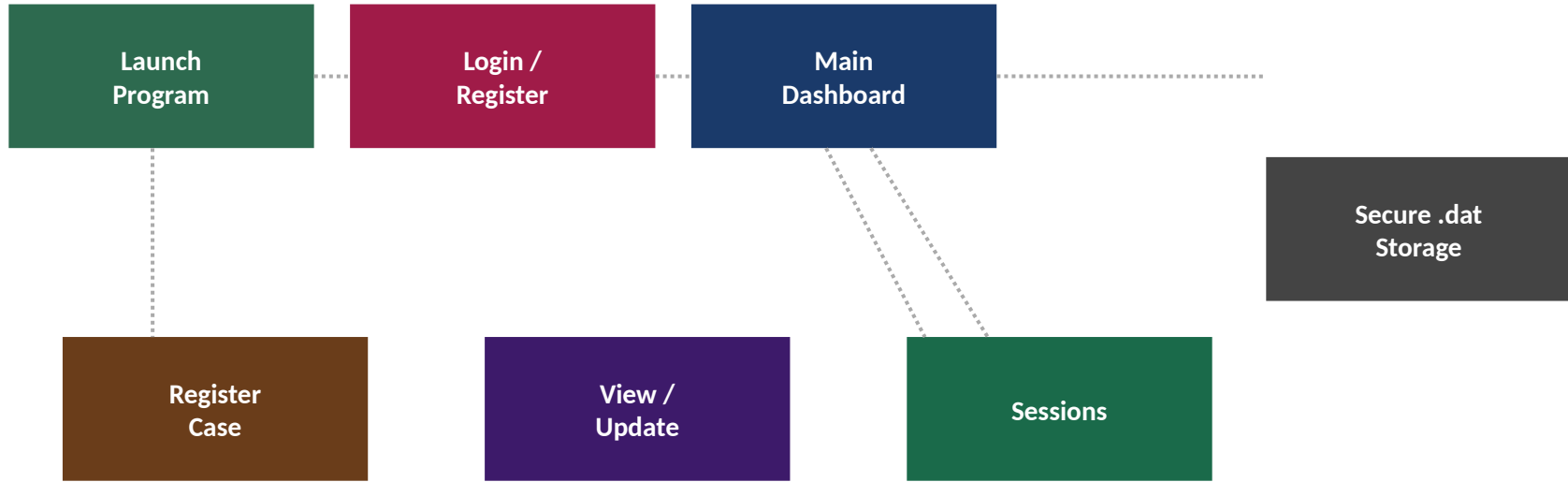
Case Registration

Unique IDs (C0001+)
Patient info + severity
Timestamped records

Session Tracking

Session IDs (S00001+)
Date + duration + notes
Linked to case ID

Working — System Flow



Conclusion

- ✓ Built a full Mental Health Case Tracking System in C++ (under 200 lines)
- ✓ Secure login & registration with FNV-1a password hashing — no plain-text passwords
- ✓ Case registration with unique IDs, severity, timestamps, and status updates
- ✓ Session tracking links therapy sessions to cases with date, duration, and notes
- ✓ All data stored persistently in .dat files — no external library needed
- ✓ Covers all deliverables: case registration, session tracking, secure data storage





ITM SKILLS
UNIVERSITY

Thank You